

The relative Berkovich boundary inclusion fails

A disk-algebra counterexample for arXiv:1410.5893

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Outcome. The final question of Leung–Ng has a negative answer, already for commutative unital complex Banach algebras. Take

$$k = \mathbb{C}, \quad A = C(\mathbb{T}), \quad a(\zeta) = \zeta, \quad B = \overline{\mathbb{C}[a]} = A(\mathbb{D})|_{\mathbb{T}}.$$

Then

$$\mu_{\mathbb{C}}(0) \in \partial \sigma_{B, \mathbb{C}}^{\text{Ber}}(a) \quad \text{but} \quad \mu_{\mathbb{C}}(0) \notin \sigma_{A, \mathbb{C}}^{\text{Ber}}(a),$$

where the boundary is taken, as in the source, in the full affine analytic line $\mathbb{A}_{\mathbb{Z}_1}^1$.

1 The exact question

For a unital k -Banach algebra A , an element $a \in A$, and the closed unital k -Banach subalgebra B generated by a , the source asks whether

$$\partial \sigma_{B, k}^{\text{Ber}}(a) \subseteq \sigma_{A, k}^{\text{Ber}}(a). \quad (1)$$

The convention established immediately before this residual question is that the boundary is taken in $\mathbb{A}_{\mathbb{Z}_1}^1$, not merely in the slice of points extending the valuation of k .

(v) For any $\mathbf{x} \in \mathbb{A}^+$, one has $\mathbb{Q}_0 \otimes_{\mathbb{Q}_0} \mathbf{x} = \mathbf{x}$ and we know from Remark 4.1(v) that $\sigma_{\mathbb{Q}_0}^{\mathbf{u}}(0) = \{0_{\mathbb{Q}}\}$. The unital Banach subring of $\mathbb{Q}_0$ generated by 0 is $\mathbb{Z}_0$. Again Proposition 4.7(b)ℰ(c) and the definition of Φ_0 tells us that $\sigma_{\mathbb{Z}_0}^{\mathbf{u}}(0) = (\mathbb{A}_{\mathbb{Z}_1}^1)_0 \cap \mathbb{A}_{\mathbb{Z}_0}^1$. The same argument as part (a) implies that $0_p^\omega \in \partial \sigma_{\mathbb{Z}_0}^{\mathbf{u}}(0) \setminus \sigma_{\mathbb{Q}_0}^{\mathbf{u}}(0)$.

Nevertheless, we do not know whether $\partial \sigma_{B, \mathfrak{k}}^{\text{Ber}}(a) \subseteq \sigma_{A, \mathfrak{k}}^{\text{Ber}}(a)$, when A is a unital $\mathfrak{k}$ -Banach algebra, $a \in A$ and B is the unital $\mathfrak{k}$ -Banach subalgebra of A generated by a . Note that a positive answer to this question will give the non-emptiness of $\sigma_{A, \mathfrak{k}}^{\text{Ber}}(a)$ (because of Propositions 4.7(a) and 4.12(d)).

Figure 1: The residual question on source PDF page 25.

That ambient-space convention is decisive: even for $k = \mathbb{C}$, the relative spectrum has transverse valuation directions arbitrarily close to each of its points.

2 Every complex-relative spectral point is a boundary point

Write $\mathbb{C}_1 = (\mathbb{C}, |\cdot|)$ and, for $0 < \nu < 1$, let $\mathbb{C}_\nu = (\mathbb{C}, |\cdot|^\nu)$. If $z \in \mathbb{C}$, denote by $\mu_{\mathbb{C}_\nu}(z) \in \mathbb{A}_{\mathbb{Z}_1}^1$ the multiplicative seminorm

$$p \longmapsto |p(z)|^\nu \quad (p \in \mathbb{Z}[t]).$$

Lemma 1. *Let E be any unital complex Banach algebra and $x \in E$. Then*

$$\partial\sigma_{E,\mathbb{C}_1}^{\text{Ber}}(x) = \sigma_{E,\mathbb{C}_1}^{\text{Ber}}(x)$$

as subsets of $\mathbb{A}_{\mathbb{Z}_1}^1$.

Proof. The relative spectrum is compact, hence closed, so its boundary is contained in it. Conversely, take ξ in the relative spectrum. By Proposition 4.13(b) of the source, $\xi = \mu_{\mathbb{C}_1}(z)$ for some z in the usual complex spectrum of x .

Choose $\nu_j \uparrow 1$ with $\nu_j < 1$. Pointwise on $\mathbb{Z}[t]$,

$$\mu_{\mathbb{C}_{\nu_j}}(z)(p) = |p(z)|^{\nu_j} \longrightarrow |p(z)| = \mu_{\mathbb{C}_1}(z)(p),$$

so $\mu_{\mathbb{C}_{\nu_j}}(z) \rightarrow \xi$ in $\mathbb{A}_{\mathbb{Z}_1}^1$. None of these approximating points belongs to a spectrum relative to $\mathbb{C}_1$. Indeed, every complete valuation-field extension of $\mathbb{C}_1$ is isometric on the scalar field and hence assigns the constant polynomial 2 the value 2, whereas

$$\mu_{\mathbb{C}_{\nu_j}}(z)(2) = 2^{\nu_j} \neq 2.$$

Thus every neighborhood of ξ meets the complement of the relative spectrum, proving that ξ is a boundary point. $\square$

3 The disk-algebra counterexample

Let $A = C(\mathbb{T})$ with the supremum norm and pointwise multiplication, and put $a(\zeta) = \zeta$. Let B be the closed unital subalgebra generated by a . By the maximum-modulus theorem, restriction to $\mathbb{T}$ identifies B isometrically with the disk algebra $A(\mathbb{D})$: it is exactly the uniform closure of analytic polynomials in the coordinate function. Thus this is precisely the generated subalgebra required in (1), with no change of norm.

Theorem 1. *For this inclusion $B \subset A$,*

$$\mu_{\mathbb{C}_1}(0) \in \partial\sigma_{B,\mathbb{C}_1}^{\text{Ber}}(a) \setminus \sigma_{A,\mathbb{C}_1}^{\text{Ber}}(a).$$

Consequently, the inclusion in (1) is false.

Proof. First compute the ordinary spectra. In B , if $|\lambda| > 1$, the Neumann series

$$(\lambda - a)^{-1} = \lambda^{-1} \sum_{n \geq 0} (a/\lambda)^n$$

converges in B . If $|\lambda| \leq 1$, evaluation at λ is a continuous unital character on $A(\mathbb{D})$ and annihilates $a - \lambda$, so $a - \lambda$ is not invertible. Hence

$$\sigma_B^{\mathbb{C}}(a) = \overline{\mathbb{D}}.$$

In $A = C(\mathbb{T})$, the function $a - \lambda$ is invertible exactly when it has no zero on $\mathbb{T}$, and therefore

$$\sigma_A^{\mathbb{C}}(a) = \mathbb{T}.$$

Proposition 4.13(b) of the source identifies the relative Berkovich spectra with the folded ordinary spectra:

$$\sigma_{B,\mathbb{C}_1}^{\text{Ber}}(a) = \mu_{\mathbb{C}_1}(\overline{\mathbb{D}}), \quad \sigma_{A,\mathbb{C}_1}^{\text{Ber}}(a) = \mu_{\mathbb{C}_1}(\mathbb{T}).$$

Thus $\mu_{\mathbb{C}_1}(0)$ belongs to the first set and not the second. Lemma 1 says that it is a boundary point of the first set in the source's ambient topology, which proves the claim. $\square$

4 Why the counterexample is robust

The failure is not caused by noncommutativity, an exotic field, a nonisometric embedding, or a potentially degenerate completed tensor product. Both algebras are familiar commutative uniform complex Banach algebras, and B is literally the inherited-norm closed subalgebra generated by one element.

The mechanism is structural. Lemma 1 shows that, with boundary taken in the full $\mathbb{A}_{\mathbb{Z}_1}^1$, (1) would force *full* spectral permanence for every complex generated subalgebra, not only the classical boundary inclusion. The disk algebra has the standard spectral hole $\mathbb{D}$, and the point 0 is the smallest explicit witness.

Bounded primary-source searches through 17 August 2026 found the source but no later arXiv paper recording this counterexample. External expert review of the ambient-boundary convention and Proposition 4.13(b) is recommended before dissemination.

References

- [1] C.-W. Leung and C.-K. Ng, *Berkovich spectra of elements in Banach Rings*, arXiv:1410.5893 (2014; manuscript dated 2018).